

Context Parallelism, from scratch

Sharding the sequence in `picotron`

ring attention (already shipping) -> **DeepSpeed-Ulysses** layered alongside it

bit-exact with a single-GPU reference, gradient-validated and benchmarked on 2x A100

What we are parallelizing

TP shards the hidden dim, SP shards the norm/residual regions, but the **attention score matrix is still $O(s^2)$ per head**. Past some sequence length, one GPU cannot hold the activations.

Context parallelism shards the **sequence itself** across a dedicated grid axis (`cp`): each rank holds `S/cp` tokens.

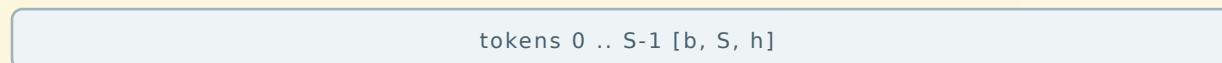
- a real axis in `DP x PP x CP x EP x TP`, with its **own process group** (`cp_group`)
- every op is per-token... **except attention**, where token `i` attends to tokens on other ranks

picotron shards the sequence contiguously in `data.py`; RoPE is sliced per shard.

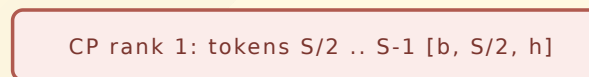
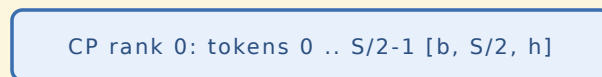
The setup: sequence sharded across CP

Context parallelism: shard the sequence across CP ranks

full sequence



split along sequence (data.py)



Every op is per-token EXCEPT attention, which needs all tokens.

RoPE is sliced per shard; only attention must communicate across the CP group.

The whole problem reduces to one question: **how do we run attention when no rank has the full sequence?** Two answers follow.

Two answers

Two ways to do attention when the sequence is sharded

send/recv K,V (P2P), step by step

Ring attention

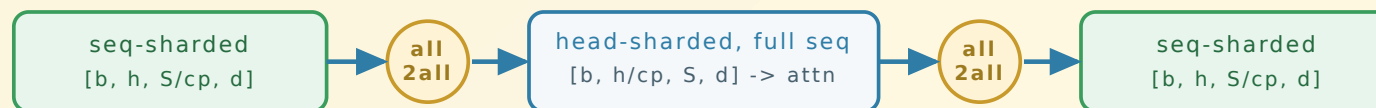
sequence stays sharded; K/V rot



online-softmax merges each block; cost $\sim O(S^2/cp)$

Ulysses attention

all-to-all to head-parallel, dense local attention, all-to-all back



Trade-off

- Ulysses: 2 all-to-alls per attention, length-independent; dense kernel composes with flash. Faster on a node.
- Ulysses limit: needs num_heads (per TP rank) divisible by cp -- cannot scale past the head count.
- Ring: P2P, no head-count limit, scales to very long sequences; flash kernel per block (FLASH_ATTEN=1).

Ring attention (`CP_ATTENTION=ring`, default)

Keep Q/K/V sharded `[b, h, S/cp, d]`. Walk the CP ring `cp` times:

1. compute attention of local Q vs the K/V block currently held
2. P2P-send K/V to the next rank, receive the next block
3. an **online softmax** merges per-block outputs -- never materialize `S x S`

Strengths

- P2P only; no all-to-all
- **no head-count limit**
- scales to very long seq / multi-node

Costs

- attention done in `cp` blocked steps
- online-softmax bookkeeping
- per-step P2P (vs Ulysses' 2 all-to-alls)
- flash kernel per block (`FLASH_ATTEN=1`); python ref otherwise

Ulysses attention (CP_ATTENTION=ulysses)

Don't move K/V -- move the **layout**. Two all-to-alls bracket an ordinary attention:

```
in   [b, num_heads,    S/cp, d]    sequence sharded
a2a  [b, num_heads/cp, S,    d]    head sharded, full seq  (scatter heads, gather seq)
attn standard dense / flash attention, this rank's heads, full sequence
a2a  [b, num_heads,    S/cp, d]    back to sequence sharded (scatter seq, gather heads)
```

Same elements reshuffled ($\text{num_heads}/\text{cp} * S == \text{num_heads} * S/\text{cp}$) -> per-rank size unchanged.
The attention is a **normal kernel**: composes with flash, no online softmax. Comm = 2 all-to-alls, **independent of sequence length**.

The backward is the same collective

An all-to-all's transpose is another all-to-all with scatter/gather dims swapped -- one tiny autograd function covers both directions:

```
class _SeqAllToAll(torch.autograd.Function):
    @staticmethod
    def forward(ctx, group, x, scatter_dim, gather_dim):
        ctx.scatter_dim, ctx.gather_dim, ctx.group = scatter_dim, gather_dim, group
        return _all_to_all(x, scatter_dim, gather_dim, group)

    @staticmethod
    def backward(ctx, grad):
        return (None, _all_to_all(grad, ctx.gather_dim, ctx.scatter_dim, ctx.group), None, None)
```

The limit: `num_heads` (per TP rank) must be divisible by `cp` -- Ulysses cannot scale CP past the head count. GQA is expanded upstream, so K/V head count is never the constraint.

Correctness (bit-exact)

```
CUDA_DEVICE_MAX_CONNECTIONS=1 torchrun --nproc_per_node 2 tests/test_cp_ulysses.py
```

Full-sequence single-GPU reference vs sequence-sharded Ulysses. Every param is replicated and each rank sees one token shard, so gradients are partial sums -> sum over the CP group recovers the reference.

check (fp32, SDPA, cp=2)	result
forward logits vs reference	max_diff = 4.8e-7
loss (CP-summed) vs reference	diff = 1.5e-5
gradients (CP-summed) vs reference	max_diff = 7.2e-7

Benchmark (2x A100, bf16, cp=2, flash in both)

seq (per rank)	attention	ms/step	tok/s	peak MB	speedup
4k (2k/rank)	ring	48.6	84,301	2,593	1.00x
4k (2k/rank)	ulysses	35.0	117,010	2,627	1.39x
8k (4k/rank)	ring	77.3	105,922	4,364	1.00x
8k (4k/rank)	ulysses	62.2	131,802	4,431	1.24x
16k (8k/rank)	ring	167.9	97,577	7,905	1.00x
16k (8k/rank)	ulysses	135.6	120,828	8,039	1.24x

Now flash-vs-flash: the earlier ~2.8x (python ring vs flash Ulysses) was an implementation artifact. The real algorithmic gap at cp=2 is ~1.2-1.4x -- Ulysses does one dense flash + 2 all-to-alls; ring does `cp` flash calls + per-step P2P, with non-zigzag causal load imbalance. Memory is now ~equal.

--compile: `torch.compile` is a modest ~3-8% win (a bit more for ring) and ~5% less peak; it does *not* change the ranking -- Ulysses keeps its ~1.2-1.4x lead. Compile only fuses the dense regions (linear/norm/rotary/MLP); the collectives + flash stay opaque. (Gotcha: the `CP_ATTENTION` env switch is constant-folded without a dynamo guard, so timing both in one process makes the 2nd reuse the 1st's graph -- `bench_cp.py` now `torch._dynamo.reset()` s per config)

Scaling to very long seq (cp=2)

seq	ring ms	uly ms	speedup	ring MB	uly MB
16k	166.6	135.7	1.23x	7,905	8,039
32k	441.7	343.8	1.28x	14,987	15,256
64k	1341.4	1003.2	1.34x	29,152	29,689
128k	4568.0	3310.3	1.38x	57,482	58,556
256k	OOM	OOM	--	--	--

Ulysses' lead *grows* with seq (1.23x -> 1.38x), it doesn't shrink. Fixed small `cp`: the `O(s^2)` term dominates and ring's causal load imbalance (rank 1 does 2 blocks) costs more; Ulysses runs one balanced dense flash.

Ring's mem edge is real but tiny (~2%, the Ulysses all-to-all transient) -- both OOM at ~256k. So at cp=2 single node there's **no seq where ring wins**. Ring's regime (`cp > num_heads`, multi-node) needs >2 GPUs to show.

How to choose

Situation	Use
Single node, <code>cp <= num_heads</code> , want speed	<code>CP_ATTENTION=ulysses</code>
<code>cp > num_heads</code> , extreme seq, multi-node	<code>CP_ATTENTION=ring</code> (default)

```
CP_ATTENTION=ulysses CUDA_DEVICE_MAX_CONNECTIONS=1 \  
torchrun --nproc_per_node 8 train.py --config cfg.yaml
```

- Ulysses: kernel-friendly, length-independent comm, capped at the head count
- Ring: P2P, no cap, the long-context / large-CP workhorse

File map & how to run

File	Role
context_parallel.py	ring attention: flash kernel (FLASH_ATTEN=1) or python ref + online softmax
ulysses.py	all-to-all head redistribution + local attention
cp_communications.py	the ring P2P send/recv primitive
tests/test_cp_ulysses.py	Ulysses vs single-GPU reference (bit-exact, fp32)
tests/test_cp_ring_flash.py	flash ring vs single-GPU flash reference (bf16)
tests/bench_cp.py	ring vs Ulysses: step time / tok/s / peak memory

```
CUDA_DEVICE_MAX_CONNECTIONS=1 torchrun --nproc_per_node 2 tests/test_cp_ulysses.py
CUDA_DEVICE_MAX_CONNECTIONS=1 torchrun --nproc_per_node 2 tests/test_cp_ring_flash.py
CUDA_DEVICE_MAX_CONNECTIONS=1 FLASH_ATTEN=1 torchrun --nproc_per_node 2 tests/bench_cp.py \
  --heads 16 --seq 8192 --hidden 2048 --inter 8192 --layers 4
```

Recap

Context parallelism = shard the sequence. Only attention needs to talk across ranks.

- **Ring:** rotate K/V, online softmax -- no head-count limit, scales far
- **Ulysses:** all-to-all to head-parallel, dense attention -- fast & kernel-friendly, capped at `num_heads`

one switch, same CP group: `CP_ATTENTION=ring | ulysses`